JEE APEX FTS – JEE ADV. FULL SYLLABUS TEST-07_[2024 P1]

KEY SHEET

MATHEMATICS

1	**A**	2	**A**	3	**A**	4	**D**	5	**ABC**	6	**CD**
7	**ABD**	8	**5**	9	**1**	10	**2**	11	**78**	12	**15**
13	**64**	14	**C**	15	**A**	16	**C**	17	**A**		

PHYSICS

18	**A**	19	**C**	20	**A**	21	**D**	22	**BC**	23	**ACD**
24	**ACD**	25	**6**	26	**2**	27	**20**	28	**4800**	29	**5**
30	**60**	31	**B**	32	**C**	33	**B**	34	**B**		

CHEMISTRY

35	**C**	36	**D**	37	**A**	38	**C**	39	**ABD**	40	**ABD**
41	**B**	42	**40**	43	**12**	44	**920**	45	**5**	46	**11**
47	**9**	48	**B**	49	**D**	50	**A**	51	**A**		

SOLUTIONS
MATHEMATICS

1. Sol. Hyperbola $2x^2-2y^2-4x+41=0$ and director circle $x^2+y^2=a^2+b^2$ meets only two points (In this case touches)

$\Rightarrow 2x^2-2\left(a^2+b^2-x^2\right)-4x+41=0$

$4x^2-4x+41-2\left(a^2+b^2\right)=0$ roots are real and equal $D=0$

$16-4\times4\left(41-2\left(a^2+b^2\right)\right)$

$a^2+b^2=20 \qquad a,b\in N$

$a=4,b=2$ or $a=2,b=4$

Hence, $|a-b|=2$

2. Sol. Let $A=(1+x)^{300},B=\left(1+x^2\right)^{200},C=\left(1+x^3\right)^{150}$

$n(A)=301,n(B)=201,n(C)=151$

$n(A\cap B)=151,n(B\cap C)=67,n(A\cap C)=101,n(A\cap B\cap C)=51$

$\therefore n(A\cup B\cup C)=385$

3. Sol. $1=\int_{2\pi/3}^{\pi}|2\cos x+1|dx=-\int_{2\pi/3}^{\pi}(2\cos x+1)dx$

$=(2\sin x+x)\Big|_{\pi}^{2\pi/3}=\sqrt{3}+\frac{2\pi}{3}-\pi=\sqrt{3}-\frac{\pi}{3}$

4. Sol. $P(B\ |S)=$ Probability that it is biased given that it shows a six

$$=\frac{P(B)P(S/B)}{P(B)P(S/B)+P(\overline{B})P(S/\overline{B})}$$

$P(B) \qquad =\frac{{}^2C_1}{{}^4C_1}=\frac{1}{2}=P(\overline{B})$

Let p be the probability that a non-multiple of 3 comes on biased die

$\therefore 4p+4p=1\Rightarrow p=\frac{1}{8}$

$$\therefore P(B/S)=\frac{\frac{1}{2}\times\frac{2}{8}}{\frac{1}{2}\times\frac{2}{8}+\frac{1}{2}\times\frac{1}{6}} \qquad =\frac{3}{5}$$

5. Sol. $\because P_1,P_2,P_3$ intersect xy plane

For point P put $z=0$

$$\left.\begin{aligned}2x+y&=1\\x-y&=2\\4x-y&=5\end{aligned}\right\}P=(1,-1,0)$$

For point $Q(yz$ plane put $x=0)$

$\left.\begin{matrix} y+z=1 \\ -y+z=2 \end{matrix}\right\} Q=\left(0,-\frac{1}{2},\frac{3}{2}\right)$ and $-y+\alpha z=5$

$\frac{1}{2}+\alpha\frac{3}{2}=5\Rightarrow\frac{3\alpha}{2}=\frac{9}{2}\Rightarrow\alpha=3$

(A) Let D.R. of line is (a,b,c)

Hence, equation of line passing through P is $\frac{x-1}{a}=\frac{y+1}{b}=\frac{z}{3}$

and DR of plane P_3 is $(4,-1,3)$

Hence, plane is $\frac{x-1}{4}=\frac{y+1}{-1}=\frac{z}{3}$

(B) Projection of $\vec{a}$ on $\vec{b}$ is $\vec{a}\cdot\hat{b}\Rightarrow$ Projection of $\overrightarrow{PQ}$ an x-axis is $\overrightarrow{PQ}\cdot\hat{i}=1$

(C) Centroid $\frac{0+1+0}{3},\frac{0-1-\frac{1}{2}}{3},\frac{0+0+\frac{3}{2}}{3}=\left(\frac{1}{3},-\frac{1}{2},\frac{1}{2}\right)$

6. Sol. $\left(\vec{a}+\vec{b}\right)\cdot\left(-7\hat{i}+2\hat{j}+3\hat{k}\right)=\pm(-14+6+12)=\pm4$

7. Sol. Since $\vec{a},\vec{b},\vec{c}$ are linearly independent $\Rightarrow\left[\vec{a}\vec{b}\vec{c}\right]\neq0$

$\Rightarrow\vec{d}$ can be written as linear combination of $\vec{a},\vec{b}$ and $\vec{c}$

$$\vec{d}=\frac{\left[\vec{d}\ \vec{b}\vec{c}\right]\vec{a}+\left[\vec{d}\vec{c}\vec{a}\right]\vec{b}+\left[\vec{d}\vec{a}\ \vec{b}\right]\vec{c}}{\left[\vec{a}\ \vec{b}\vec{c}\right]}$$

Now, $\vec{r}=\left[\vec{d}\vec{b}\vec{a}\right]\vec{c}-\left[\vec{d}\vec{b}\vec{c}\right]\vec{a}+\left[\vec{d}\vec{c}\vec{b}\right]\vec{a}-\left[\vec{d}\vec{c}\vec{a}\right]\vec{b}+\left[\vec{d}\vec{a}\vec{c}\right]\vec{b}-\left[\vec{d}\vec{a}\vec{b}\right]\vec{c}$

$=2\left\{\left[\vec{d}\vec{c}\ \vec{b}\right]\vec{a}+\left[\vec{d}\vec{a}\vec{c}\right]\vec{b}+\left[\vec{d}\ \vec{b}\vec{a}\right]\vec{c}\right\}=-2\left[\vec{a}\ \vec{b}\vec{c}\right]\vec{d}\Rightarrow\vec{r}=-2\left[\vec{a}\ \vec{b}\vec{c}\right]\vec{d}$

8. Sol. Let $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ solve, $a+b+c+d=5$

$a=-1,b=0,c=4,d=2$

9. Sol. Conceptual

10. Sol. Put $x=t^2$

11. Sol.

$$\left|\vec{a}-\vec{b}\right|^2+\left|\vec{b}-\vec{c}\right|^2+\left|\vec{c}-\vec{a}\right|^2=2\left(\left|\vec{a}\right|^2+\left|\vec{b}\right|^2+\left|\vec{c}\right|^2\right)-2\left(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}\right)$$

$=52-2\left(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}\right)\leq78$ Because, $\left|\vec{a}+\vec{b}+\vec{c}\right|^2\geq0$

$\Rightarrow\left|\vec{a}\right|^2+\left|\vec{b}\right|^2+\left|\vec{c}\right|^2+2\left(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}\right)\geq0$ $\Rightarrow2\left(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}\right)\geq-26$

12. Sol. $f(x)=(x-2)g(x)+13$

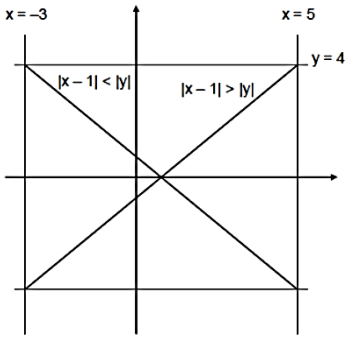
Let α be integral root then $f(\alpha)=0$

$\therefore (\alpha-2)g(\alpha)+13=0 \Rightarrow g(\alpha)=-\dfrac{13}{\alpha-2}$ as $g(\alpha)$ is integral, so $\alpha=1,3,15,-11$

Same $f(x)=(x-10)h(x)+5$ and $f(\alpha)=0 \Rightarrow h(\alpha)=-\dfrac{5}{\alpha-10} \Rightarrow \alpha=15$

So, α satisfy both that is 15

13. Sol. Required area $=8\times 8=64$



14. Sol. (P) Continuity must be checked at $x=1,e,3,\sqrt{10},\sqrt{11},\sqrt{12},3.5$ Further, $f(1)=0$ and $\lim\limits_{x\to 1} f(x)=\lim\limits_{x},\operatorname{sgn}(x-2)\times[\log x]=0$

Hence, $f(x)$ is continuous at $x=1$

$f(e)=1, f(e^{-})=0$

Hence $f(x)$ is discontinuous at $x=e$

$\lim\limits_{x\to 3^{+}} f(x)=\lim\limits_{x\to 3^{-}}\operatorname{sgn}(x-2)\times[\log_e x]=1 \qquad \Rightarrow \lim\limits_{x\to 3^{+}} f(x)=\lim\limits_{x\to 3^{+}}\{x^2\}=0$

Hence, $f(x)$ is discontinuous at $x=3$

Also $\{x^2\}$ is discontinuous $x=\sqrt{10},\sqrt{11},\sqrt{12}$ therefore,

$\lim\limits_{x\to \frac{7^{-}}{2}} f(x)=\lim\limits_{x\to 3^{-}}\{x^2\}=0.25=f(3.5)$

Hence, $f(x)$ is discontinuous at

$x=3,\sqrt{10},\sqrt{11},\sqrt{12},e$

(Q) clearly there is one point of inflexion

(R) not differentiable at $x=1$ and $x=\sqrt{3}$

15. Sol. (A) $m=\dfrac{1}{t}$ points are (at^2, 2at), $\left(\dfrac{a}{t^2},\dfrac{-2a}{t}\right)$

(B) Semi latus rectum is H.M of PS and P'S of focal chord of ellipse

$\dfrac{2(SP')\times 4}{SP'+4}=\dfrac{4}{3} \qquad 6SP'=SP'+4 \Rightarrow SP'=\dfrac{4}{5}$

(C) $\perp^{r}$ asymptotes $\Rightarrow e=\sqrt{2}$

(D) $r=\sqrt{3}$

16. Sol. (A) We must have $e_1 < 1 < e_2 \Rightarrow f(1) < 0$ (location of roots)

$= 1 - a + 2 \langle 0 \Rightarrow a \rangle 3$

(B) We must have both the roots greater than 1

I. $D \geq 0$ or $a^2 - 8 \geq 0$ or $a \in \left(-\infty, -2\sqrt{2}\right] \cup \left[2\sqrt{2}, \infty\right)$

II. $f(1) > 0$ or $1 - a + 2 > 0$ or $a < 3$

III. $\frac{a}{2} > 1 \Rightarrow a > 2$

From equation (i), (ii) and (iii), we have $a \in [2\sqrt{2}, 3)$.

(C) We must have $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$

$\Rightarrow \frac{(e_1 + e_2)^2 - 2e_1e_2}{e_1^2 e_2^2} = 1 \qquad \Rightarrow \frac{a^2 - 4}{4} = 1 \qquad \Rightarrow a = \pm 2\sqrt{2}$

(D) We must have $e_1 < \sqrt{2} < e_2$

17. Sol. Let the plane be $\alpha x + \beta z + 1 = 0$.

Passes through $(1,0,1)$ & $(3,2,-1)$ $\quad \therefore \alpha = -\frac{1}{2}, \beta = -\frac{1}{2} \therefore (P)\, x + z = 2$

(Q) (Q) Both A & B are same side of P Reflection of A in plane is $A'(2,0,-2)$ equation of line

(S) Also, A' will lie on $\frac{x-2}{1} = \frac{y-\alpha}{0} = \frac{z+B}{-1}$

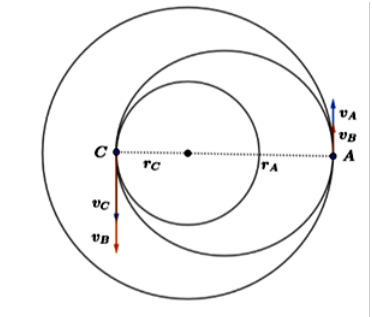
$\Rightarrow \frac{2-2}{1} = \frac{0-\alpha}{0} = \frac{-2+\beta}{-1}$

$\Rightarrow \alpha = 0, \beta = 2$

$\alpha^4 + \beta^4 = 16$

PHYSICS

18. Sol.



Recall the velocity of a circular orbit is

$v = \sqrt{\frac{GM}{r}}$ The angular momentum is then $L = mvr = m\sqrt{GMr}$

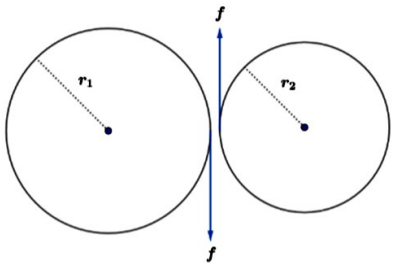
Since $r_A > r_C$, we have $L_A > L_C$. At the apogee, the velocity v_B of satellite B is smaller than the velocity v_A of satellite A since the ellipse is contained in the A circle. Thus, we

have $L_B < L_A$. Similarly, we find $v_B > v_C$ at the perigee since the ellipse contains the C circle. Thus, $L_B > L_C$. Finally, we obtain $L_A > L_B > L_C$ so the answer is A.

19. Sol. Conceptual

20. Sol. The gauge pressure of the water directly under block m_1 must be $\frac{m_1 g}{A_1}$, where A_1 is the cross-sectional area of the tube containing block m_1. The pressure in the horizontal section of the tube is therefore $pgh + \frac{m_1 g}{A_1}$. But by the same reasoning, the pressure is also $pgh + \frac{m_2 g}{A_2}$ and $pgh + \frac{m_2 g}{A_3}$. We conclude $\frac{m_1}{A_1} = \frac{m_2}{A_2} = \frac{m_3}{A_3}$. This is the condition for equilibrium in the device. It we close valve A, the condition for equilibrium in the right two tubes is $\frac{m_2}{A_2} = \frac{m_3}{A_3}$, which is already met, so these tubes remain in equilibrium. The same applies to valve *B*. In other words, the pressure is the same to the left and right of a valve, so closing the valve doesn't affect the equilibrium.

21. Sol.



Since both discs come to rest, they go to rest at the same time. From angular kinematics, $\omega - \alpha t = 0\ t = \frac{\omega}{\alpha}$ so the quantity ω / α is the same for both disks:

$\frac{\omega_1}{\alpha_1} = \frac{\omega_2}{\alpha_2}$ Let's rewrite α in terms of r. We have $\alpha = \frac{\tau}{I} \propto \frac{r}{mr^2} \propto \frac{r}{r^4} = \frac{1}{r^3}$

using the fact that $\tau \propto r$ as the same force (friction) acts on both disks and also $m \propto r^2$ as both disks have the same density and thickness. Substituting into the earlier equation $\omega_1 r_1^3 = \omega_2 r_2^3$,

22. In case of perfectly elastic collision, if $m_1 = m_2$ then velocities are interchanged.
In case of perfectly Inelastic collision, apply conservation of linear momentum

23. Potential due to non conducting circular disc at a distance of x from its centre is $\frac{\sigma}{2\varepsilon_0}\left(\sqrt{R^2 + x^2} - x\right)$

So, potential at a distance of $\sqrt{3}R$ due to the given portion is

$$V_P = \frac{\sigma}{4\varepsilon_0}\left(\sqrt{R^2 + 3R^2} - \sqrt{3}R\right) + \frac{\sigma}{4\varepsilon_0}\left(\sqrt{4R^2 + 3R^2} - \sqrt{3}R\right) = \frac{\sigma R}{4\varepsilon_0} - 2 + \sqrt{7} - 2\sqrt{3}\Bigg]$$

Potential at the centre due to the given portion is

$$V_C = \left(\frac{\sigma \cdot 2R}{2\varepsilon_0}\right)\left(\frac{45^\circ}{360^\circ}\right)\times 4 + \left(\frac{\sigma \cdot R}{2\varepsilon_0}\right)\left(\frac{45^\circ}{360^\circ}\right)\times 4 = \frac{3}{4}\frac{\sigma R}{\varepsilon_0}$$

Work done by external agent $= \frac{3}{4}\frac{\sigma R q}{\varepsilon_0}$

24. Sol. $\Delta Q_{AB} = nC_p\Delta T = 2\times\frac{5R}{2}(900-300) = 3000\text{R}$

In the process $BC, P = \alpha T^2$

$PV^2 = \text{constant}$

Molar heat capacity in the process BC,

$$C = C_V + \frac{R}{(1-x)} = \frac{3R}{2} + \frac{R}{(1-2)} \quad C = \frac{R}{2}$$

$$\Delta Q_{BC} = nC\Delta T = 2\times\frac{R}{2}(300-900) = -600R$$

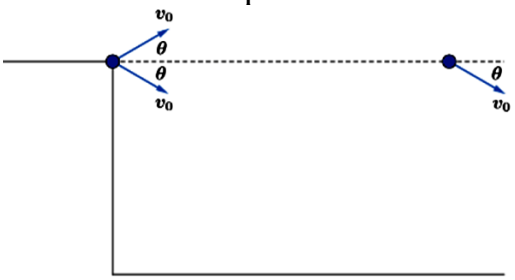
$$\Delta Q_{CA} = \Delta W_{CA} = nRT\,/\,n\left(\frac{P_C}{P_A}\right) = 2R\times 300\ln\left(\frac{1}{9}\right) = -1200R\ell n(3)$$

$$\Delta Q_{CA} = -1320R$$

$$\Delta W_{\text{cyde}} = \Delta Q_{\text{cyde}} = 3000R - 600R - 1320R = 1080R$$

$$E_{\text{Efficiency of cycle,}}\eta = \frac{\Delta W_{\text{cyde}}}{\Delta Q_{\text{suppled}}}\times 100 \qquad \eta = \frac{1080R}{3000R}\times 100 = 36\%$$

25. Recall from projectile motion that the y-component of a particle's velocity flips direction when the particle returns to its starting height.



Applied to our case, the particle thrown upwards returns to the height of the ledge with the same velocity as the particle thrown downwards. Thus, the difference in time between the two trajectories only comes from the time the particle thrown up takes to come back down. From kinematics, the time taken to reach the peak height is

$$v_y - gt = 0$$

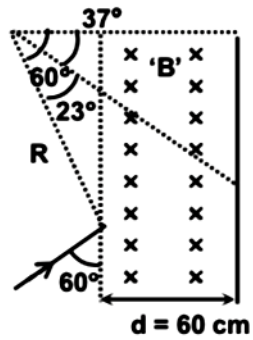
$$t = \frac{v_y}{g} = \frac{v_0\sin\theta}{g}$$

Coming back down takes the same time so the total time is

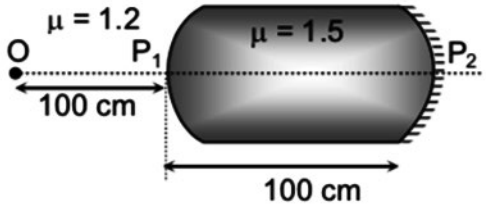
$$T = 2t = \frac{2v_0\sin\theta}{g} = \frac{2(50\text{ m/s})\sin(37^\circ)}{10\text{ m/s}^2} = 6\text{ s}$$

26. Sol. $d = R\cos 37^\circ - R\cos 60^\circ$

$R = 2m$



27. Sol. Refraction from P_1



$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

$$\frac{1.5}{v_1}-\frac{1.2}{-100}=\frac{1.5-1.2}{+20}$$

$$\Rightarrow v_1=500 \text{ cm}$$

Reflection from P_2

$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$$

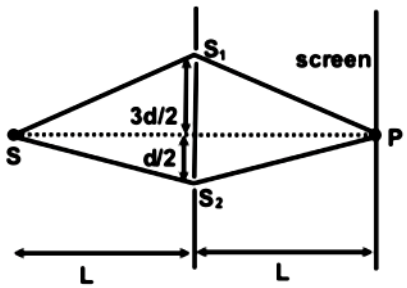
$$\frac{1}{-200}=\frac{1}{v_2}+\frac{1}{400},\ v_2=\frac{-400}{3} \text{ cm}$$

Again refraction from P_1

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R},\frac{1.2}{v_3}-\frac{1.5}{100/3}=\frac{1.2-1.5}{-20}$$

$$\Rightarrow v_3=+20 \text{ cm}(\text{ left to } P_1)$$

28. Sol. $\frac{3}{4}I_{\max}=I_{\max}\cos^2\frac{\phi}{2}\Rightarrow\cos\frac{\phi}{2}=\frac{\sqrt{3}}{2}$



$$\Rightarrow\frac{\phi}{2}=\frac{\pi}{6}\Rightarrow\phi=\pi/3 \qquad \phi=\frac{2\pi}{\lambda}\Delta x=\frac{\pi}{3}\Rightarrow\Delta x=\frac{\lambda}{6}$$

Given $\mathrm{d}=200\mu\,\mathrm{m}=2\times10^{-4}\,\mathrm{m}$

$$2\left[\sqrt{L^2+\frac{9d^2}{4}}-\sqrt{L^2+\frac{d^2}{4}}\right]=\frac{\lambda}{6}\Rightarrow\frac{2d^2}{L}=\frac{\lambda}{6}\Rightarrow\lambda=\frac{12d^2}{L}=\frac{12\times4\times10^{-8}}{1}=4800\text{Å}$$

29. Sol. The intensity of partially polarized light that is passed through a linear polarizer is $\frac{I_u}{2}+I_p\cos^2\theta$ where I_u is the intensity of the unpolarized light, I_p is the intensity of the polarized light, and θ is the angle between the polarization of the polarized light and the linear polarizer. The second term follows directly from Malus' law, and the first term is an average of Malus' law over all angles.

The maximum intensity occurs when $\theta=0^\circ$ or 180° and is given by $I_{\max}=\frac{I_u}{2}+I_p$

Per the problem statement, a 30° rotation reduces the intensity by 10%, so

$$0.9\left(\frac{I_u}{2}+I_p\right)=\frac{I_u}{2}+I_p\cos^2 30^\circ=\frac{I_u}{2}+\frac{3I_p}{4}$$

Solving for I_u gives us $I_u=3I_p$. Thus, we have $I_{\max}=\frac{5}{2}I_p,\ I_{\min}=\frac{3}{2}I_p$

where the latter occurs at $\theta=\pm90^\circ$, so $V=1/4$.

30. Sol. Take torque about bottom most point and consider direction of angular momentum of apparatus

31. Sol. For any case, Buoyancy $\vec{B}=-V\rho_{\text{liquid}}\vec{g}_{\text{eff}}$ where $\vec{g}_{\text{eff}}=\vec{g}-\vec{a}$ and $\vec{T}+\vec{B}+m\vec{g}=ma$

32. Sol. The magnetic pressure exerted on the wall of outer cylinder.

I) $P_0=\frac{I}{4\pi a}\left(\frac{\mu_0 I}{4\pi a}-\frac{\mu_0 I}{8\pi a}\right)=\frac{\mu_0 I^2}{32\pi^2 a^2}$

II) $P_0=\frac{2I}{4\pi a}\left(\frac{\mu_0 2I}{4\pi a}-\frac{\mu_0 2I}{8\pi a}\right)=\frac{\mu_0 I^2}{8\pi^2 a^2}$

III) $P_0=\frac{2I}{4\pi a}\left(\frac{\mu_0 I}{8\pi a}+\frac{\mu_0 2I}{8\pi a}\right)=\frac{3\mu_0 I^2}{16\pi^2 a^2}$

IV) $P_0=\frac{3I}{4\pi a}\left(\frac{\mu_0 3I}{8\pi a}+\frac{\mu_0 3I}{8\pi a}\right)=\frac{9\mu_0 I^2}{16\pi^2 a^2}$

33. Sol. (I) $\Delta W_{\text{cycle}}=\frac{1}{2}\times V_0\times P_0=\frac{P_0\ V_0}{2}$

(II) $\Delta Q_{AB}=nCC_P\Delta T=\frac{5}{2}nR\Delta T=\frac{5}{2}P_0\ V_0$

$$\Delta Q_{BC}=nC\Delta T=n\frac{R}{2}(T_0-2T_0)=-\frac{nRT_0}{2}=-\frac{P_0\ V_0}{2}$$

$$\Delta Q_{CA}=nRT_0\ell n\left(\frac{V_0}{4V_0}\right)=-2nRT_0\ln2=-2P_0\ V_0\ln2$$

$$\Delta W_{\text{cycle}}=\frac{5P_0\ V_0}{2}-\frac{P_0\ V_0}{2}-2P_0\ V_0\ell n2=2P_0\ V_0(1-\ell n2)$$

(III) $\Delta Q_{AB} = nC\Delta T = n\frac{3R}{2}\left(\frac{T_0}{2} - T_0\right) = -\frac{3}{4}nRT_0 = -\frac{3}{4}\times 4P_0V_0 = -3P_0V_0$

$\Delta Q_{BC} = nC_v\Delta T = n\frac{5R}{2}\left(\frac{T_0}{4} - \frac{T_0}{2}\right) = -\frac{5}{2}\frac{nRT_0}{4} = -\frac{5P_0V_0}{2}$

$\Delta Q_{CA} = nC\Delta T = n2R\Delta T = 2nR\left(T_0 - \frac{T_0}{4}\right) = \frac{6nRT_0}{4} = 6P_0V_0$

$\Delta W_{ccle} = \Delta Q_{ocde} = 6P_0V_0 - 3P_0V_0 - \frac{5P_0V_0}{2} = \frac{P_0V_0}{2}$

(IV) $\Delta Q_{AB} = nC_P\Delta T = \frac{7}{2}nR\Delta T = \frac{7}{2}P_0V_0$

$\Delta Q_{BC} = nR2T_0\ell n2 = 2P_0V_0\ell n2$

$\Delta Q_{CD} = nC_v\Delta T = n\frac{5R}{2}(T_0 - 2T_0) = -\frac{5}{2}nRT_0 = -\frac{5}{2}P_0V_0$

$\Delta Q_{DA} = nRT_0\ell n\left(\frac{1}{4}\right) = -2nRT_0\ln 2 = -2P_0V_0\ell n2$

$\Delta W_{orde} = \Delta Q_{crde} = \frac{7P_0V_0}{2} - \frac{5P_0V_0}{2} = P_0V_0$

34. Sol. $E_1 = \frac{GMm\left(3R^2 - r_L^2\right)}{2R^3} - \frac{GMm}{r_0}$

$E_2 = \frac{GMm}{2r_0}$

CHEMISTRY

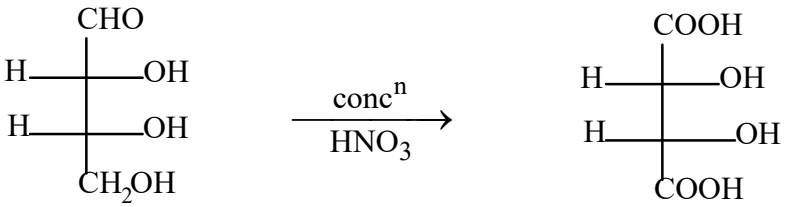
35. Sol. Most likely pKa value of aspirin is 3.5

36. Sol. $Mn^{2+} \longrightarrow MnO_4^- \quad E^\circ = -1.51V$

$PbO_2 \longrightarrow Pb^{+2} \quad \underline{E^\circ = +1.455}$

$\underline{\quad -vc \quad}$

37. Sol. Factual

38. Sol. 

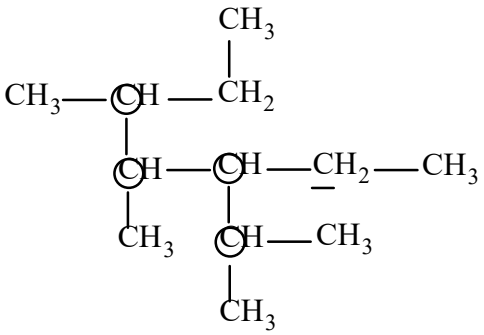
39. Sol. $NH_4I(s) \rightleftharpoons \underset{'x'}{NH_3(g)} + \underset{'x'}{HI(g)}$

$K_p = \frac{1}{4} = \left(P_{NH_3}\right)\left(P_{HI}\right) \qquad P_{NH_3} = \frac{1}{2}\text{atm} = P_{HI}$

$\left(\frac{1}{2}\right)\left(\cancel{100}\right) = (x)\left(\frac{1}{12}\times\cancel{300}\right)^3 \qquad x = \frac{12}{6} = 2\ moles$

$n_{NH_3} = n_{HI} = 2\ moles\ of\ eq^m$

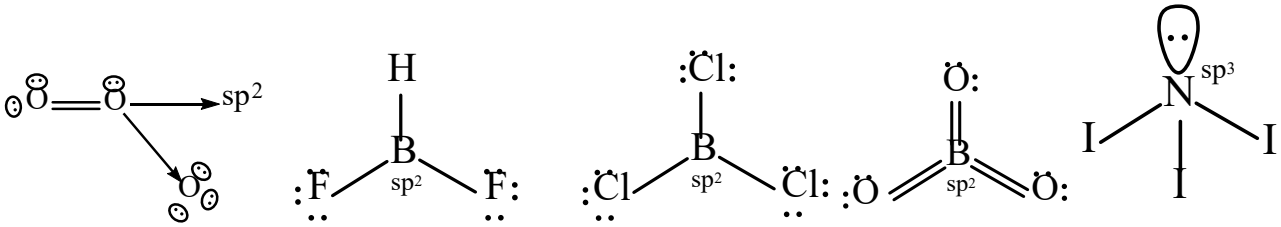
40. Sol



Ⓒ → Secondary carbon

C → Tertiary carbon

41. Sol.



42. Sol.

$$\boxed{t=\frac{2.303}{k}\log\left(\frac{P_i}{P_i-x}\right)}$$

x	$\longrightarrow$	$y+$	$2z$
P_i		P_1	P_2
P_i-x		P_1+x	P_2+2x
O		P_1+P_i	P_2+2P_i

$$\boxed{\begin{aligned} P_i+P_1+\ P_2\ &=100\\ P_i+P_1+P_2+2x&=220\\ P_1+P_2+\ 3P_i\ &=250\end{aligned}}$$

$$P_i=100-P_1-P_2$$

$$(P_i-x)=P_i \qquad \begin{pmatrix}P_i=75\\ x=60\end{pmatrix}$$

On solving

$P_i=75$

$x=60$

$$K=\frac{2.303}{20}\log\left(\frac{75}{15}\right)$$

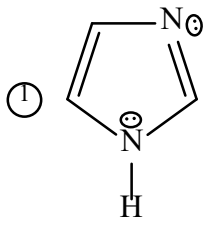
$$K=\frac{2.303}{20}\log(5)$$

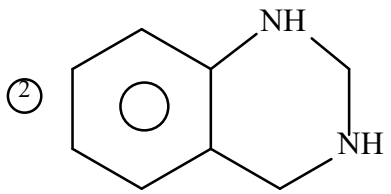
$$t_{24/25}=\frac{\cancel{2.303}\times 20}{\cancel{2.303}\log(5)}\log\left(\frac{P_i}{P_i/25}\right)$$

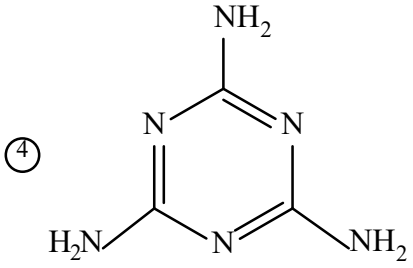
$$t_{24/25} = \frac{20}{\log(5)} \times \log(25)$$

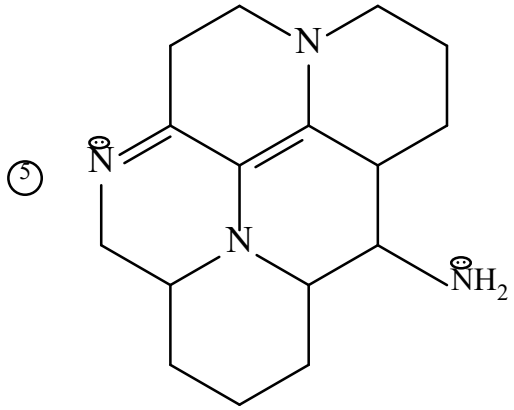
$$t_{24/25} = 40$$

43. Sol.









$1 + 2 + 4 + 5 = 12$

44. Sol. $CaO + 3C \longrightarrow CaC_2 + CO$

'x' 'y' gm of CaO reacted.

Wt. of CaO remain $= (x - y)gm$

$\underset{(x/56)}{CaO} \longrightarrow$ moles of CaO initially present

$\underset{moles}{(y/56)}$ CaO reacted

$y/56$ moles of CaC_2 formed.

Wt. of CaC_2 produced $= \left(\frac{y}{56}\right) \times 64 = \left(\frac{8y}{7}\right)$

Total wt. $= (x - y) + \left(\frac{8y}{7}\right) = x + \frac{8y}{7} - y$

$$= \left(x + \frac{y}{7}\right)gm$$

$\%$ of $CaC_2(by\ wt) = \frac{8y/7}{x + y/7} \times 100 = 64$

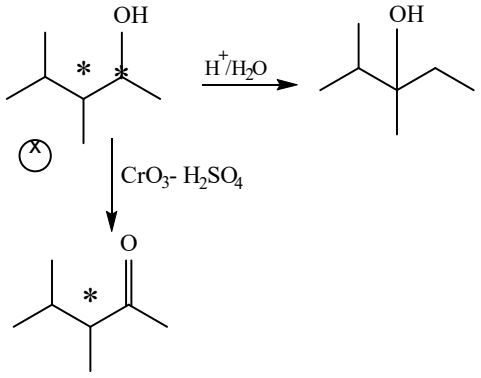
$x + y/7 = 100$ $\qquad \frac{8y}{7x + y} = \frac{64}{100}$

$8y = 0.64(7x + y)$ $\qquad 8y = 4.48x + 0.64y$

$y = \left(\frac{4.48}{7.36}\right)x$ $\qquad x + \frac{y}{7} = 1000$

$$x+\left(\frac{4.48}{7.36\times 7}\right)x=1000 \qquad (1.0869)x=1000 \quad x=920\,kg$$

45. Sol.



46. Sol.

1	0	0
HA	$\rightleftharpoons H^+$	$+\ A^-$
$c-c\propto$	$c\propto$	$c\propto$

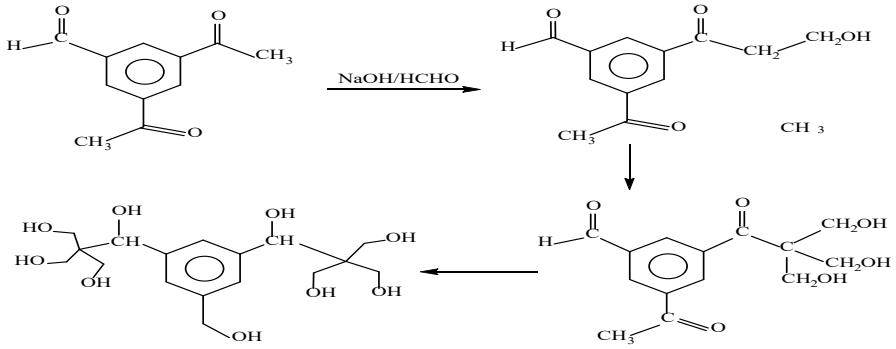
$$[H^+]=10^{-2}=c\propto \qquad 10^{-2}=(10^{-1})\propto \qquad \propto=0.1$$

$$\pi=(i)\,cRT$$

$$\pi=(1.1)(0.1)(RT)$$

$$=(0.11)\,(RT) \qquad x=0.11$$

47. Sol.



48. Sol. $K_4[Fe(CN)_6]$ with $Cu^{+2}\rightarrow Brown\ ppt.$

$Ca^{+2}\rightarrow White\ ppt$

$Fe^{+2}\rightarrow White\ ppt$

$Fe^{+2}\rightarrow White\ ppt$

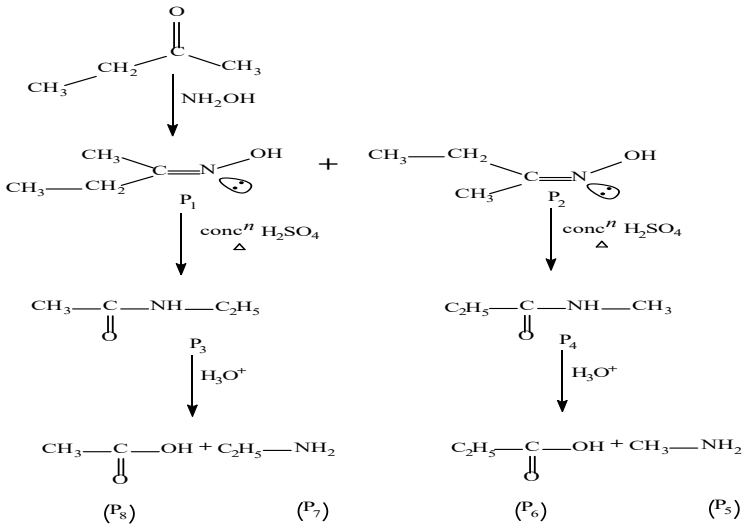
49. Sol. Brass $\longrightarrow {}_{29}Cu$ & ${}_{30}Zn$

Bronze $\longrightarrow {}_{29}Cu$ & ${}_{30}Sn$

German silver $\longrightarrow {}_{29}Cu,\ {}_{28}Ni,\ {}_{30}Zn$

Silver Amalgam $\longrightarrow {}_{47}Ag,\ {}_{80}Hg$

50. Sol.



51. Sol.

$Br_2(\ell)$	$\Delta H_f^\circ = O$	$\Delta S_f^\circ = O$
$O_3(g)$	$\Delta H_f^\circ > O$	$\Delta S_f^\circ < O$
$I_2(g)$	$\Delta H_f^\circ > O$	$\Delta S_f^\circ > O$
$CO(g)$	$\Delta H_f^\circ < O$	$\Delta S_f^\circ > O$